

PATHAHEAD · FREE ROADMAP

Become a Data Scientist / Analyst from Scratch

Turn raw numbers into decisions. Go from zero to job-ready analytics in about five months — built around real datasets and portfolio-worthy reports.

20

weeks · ~10–12 hrs/wk

5

portfolio projects

4

phases · data to insight

₹0

free resources only

A free PathAhead learning resource · adjust the pace to your life — the order matters more than the dates.

THE ONE RULE

Analyze → visualize → explain

Anyone can run a chart. The skill is explaining what it means. Every project below ends in a clear insight you could present to a manager.

THE ROADMAP

Four phases, step by step

0

Foundations

Weeks 1–4

The core toolkit: Python, stats, and SQL.

WEEK 1

Python basics

Variables, loops, functions, working with files.

WEEK 2

pandas & NumPy

DataFrames, cleaning, filtering, groupby.

WEEK 3

Statistics

Descriptive stats, distributions, probability.

WEEK 4

SQL

SELECT, joins, aggregation, grouping.

1

Analysis & Visualization

Weeks 5–9

Explore data and tell its story with charts.

WEEK 5

EDA

Explore, clean, and handle missing data.

WEEK 6

Visualization

matplotlib and seaborn for clear charts.

WEEK 7

Dashboards

Plotly, Power BI, or Tableau basics.

WEEK 8

Data storytelling

Turning findings into a clear narrative.

WEEK 9 · PROJECT ►

Analytics report

A full EDA writeup on a real dataset.

2

Statistics & ML

Weeks 10–15

Test hypotheses and build predictive models.

WEEK 10**Hypothesis testing**

A/B tests, p-values, significance.

WEEK 11**Regression**

Linear and logistic regression.

WEEK 12**Classification**

Decision trees and random forests.

WEEK 13**Clustering**

K-means and customer segmentation.

WEEK 14**Model evaluation**

Metrics, cross-validation, avoiding overfit.

WEEK 15 · PROJECT ►**Predictive model**

Predict churn or price and evaluate it.

3

Tools & Delivery

Weeks 16–20

The workflow that makes you hireable.

WEEK 16**Advanced SQL**

Window functions, CTEs, subqueries.

WEEK 17**Spreadsheet mastery**

Pivot tables, lookups, dashboards in Excel/Sheets.

WEEK 18**Git & notebooks**

Reproducible, version-controlled analysis.

WEEK 19**Communicating results**

Reports and decks for non-technical people.

WEEK 20 · CAPSTONE ►**End-to-end analysis**

A full case study from raw data to recommendation.

BUILD A PORTFOLIO

Five projects that get you hired

Projects beat certificates. Do them in order — each builds on the last. Put every one on GitHub with a clear README: problem → approach → result.

Start here

1 • Exploratory Data Analysis

Pick any Kaggle dataset. Clean it, chart it, find 3 insights.

Dashboards

2 • Interactive Dashboard

Sales, sports, or COVID data in Power BI / Tableau / Plotly.

Statistics

3 • A/B Test Analysis

Analyze an experiment and state whether the result is significant.

In demand

4 • Predictive Model

Predict churn or house prices; explain accuracy and drivers.

Capstone

5 • End-to-End Case Study

Raw data → analysis → model → written recommendation.

ROLES YOU CAN APPLY FOR

Data Analyst · Data Scientist · Business Analyst · BI Analyst

TYPICAL STARTING SALARY · INDIA

₹4–12 LPA

LEARN IT

Learning resources

Every item links to the real source (clickable in this PDF).

BOOKS

- [Python for Data Analysis](#) by Wes McKinney
- [An Introduction to Statistical Learning](#) by James, Witten, Hastie and Tibshirani
- [The Art of Statistics](#) by David Spiegelhalter

COURSES

- [Data Analysis with Python](#) by freeCodeCamp (free)
- [IBM Data Science Professional Certificate](#) by Coursera
- [Data Science](#) by NPTEL (free)
- [Google Data Analytics Certificate](#) by Coursera

YOUTUBE

- [Ken Jee](#)
- [StatQuest with Josh Starmer](#)
- [Data School](#)

PRACTICE

- [Kaggle](#)
- [DataCamp](#)
- [HackerRank](#)

DO THIS TODAY

Your first step (takes 30 minutes)

1

Open Google Colab

colab.research.google.com — free, browser-based, every library ready.

2

Get a Kaggle account

kaggle.com — thousands of free datasets and community notebooks.

3

Do one EDA

Load a dataset, run `.describe()`, plot 3 charts. You've started.

Job-ready checklist: 3–5 solid projects on GitHub (≥ 1 deployed or documented) · can explain your work in an interview · comfortable with the core tools of the field · a portfolio or profile recruiters can find.

PathAhead — India's premium career guidance platform. This roadmap is a free educational resource; timelines are a guide and everyone learns at a different pace. Referenced third-party resources (Kaggle Learn, freeCodeCamp, Khan Academy, Mode SQL Tutorial, StatQuest (YouTube)) are for learning only and are not affiliated with PathAhead.